

David Adam Nieves

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EDUCATION

Master of Science, Computer Science, University of Illinois Urbana-Champaign

5/2025

- Cumulative GPA 3.8/4.0
- Honors: Magna cum laude

Bachelor of Science, Computer Science, University of Southern New Hampshire

4/2023

- Cumulative GPA: 3.9/4.0
- Honors: Summa Cum Laude, Dean's List (all semesters)

SUMMARY OF TECHNICAL SKILLS

- Languages: Python, JavaScript, TypeScript, C++, SQL, Java, HTML, CSS, Shell, XML Systems: Microsoft Windows, Linux
- Frameworks: TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost, Flask, Django, Angular, React, Node.js, Express.js, Android SDK
- Tools: Git, GitHub, GitHub Actions, GitLab, Docker, MongoDB, PostgreSQL, MySQL, Firebase, Postman, Jest, ROS, OpenCV, Heroku, Logstash, Google Analytics
- Visualizations: Tableau, Power BI, Matplotlib, Plotly, Seaborn, Looker Studio, Dash, D3.js, Altair, Google Charts, Chart.js
- Cloud: AWS (Lambda, S3, API Gateway, CodePipeline, CloudFormation), Firebase (Authentication, Firestore, Cloud Functions), Google Cloud Platform (Cloud Functions, Cloud Build), Azure (Azure Functions, Azure DevOps), Heroku, Netlify, Vercel, Docker, Kubernetes
- IDEs and OS: Visual Studio, Eclipse, Android Studio, Microsoft Windows, Linux JSP/Servlets, JQuery, PHP

Experience:

Lowe's Software Engineer

2019-2023

- Analyze large datasets related to sales, inventory, customer behavior, operations, etc.
- Create dashboards and automated reports (commonly in Tableau, Power BI, or Excel).
- Track KPIs like revenue, conversion rates, foot traffic, or supply chain efficiency.
- Created A/B testing dashboards to support pricing and promotional strategies using tools like Excel, Python (Pandas), and Looker Studio.
- Developed internal web tools using Flask and React to allow non-technical teams to query and visualize data insights dynamically.
- Designed and implemented RESTful APIs for internal data services, improving cross-team data access and integration.
- Integrated data validation checks and logging mechanisms into ETL pipelines using **Pandas**, **Airflow**, and **Docker**.

Certificates

- Deep Learning Specialization (Coursera – Andrew Ng)
- TensorFlow Developer Certificate
- Natural Language Processing in TensorFlow
- Time Series Forecasting
- Machine Learning (Stanford University)

Technical Projects

Full-Stack User Dashboard & Analytics Platform

Angular, Node.js, Express, MongoDB, Chart.js

- Built a responsive web application for user account management, data tracking, and real-time dashboard analytics
- Implemented JWT-based authentication, RESTful APIs, and dynamic frontend components
- Designed MongoDB schema and integrated Chart.js for interactive data visualization
- Deployed with Docker and set up CI/CD using GitHub Actions
- Collaborated with team members on GitHub and deployed the application using Docker

Movie Recommendation System

Python, Flask, Scikit-learn, Surprise

- Built a hybrid recommender system using collaborative and content-based filtering
- Evaluated with RMSE and precision@k; deployed via Flask for interactive web demo

Image Classification with CNNs

TensorFlow, Keras

- Built a convolutional neural network for CIFAR-10, achieving 90%+ accuracy
- Used data augmentation and dropout to enhance generalization and robustness

Android AI Weather App

- Java, Android Studio, OpenWeatherMap API, Gemini AI, Firebase
- Collaborated with a team of 3 to develop a personalized weather app with city-based forecasts, map navigation, and AI-generated weather insights
- Integrated Gemini AI to generate dynamic, context-aware questions and responses based on live weather data
- Implemented Firebase for secure user authentication, storage of user preferences, and real-time updates
- Designed multiple custom UI layouts to reflect user settings such as location, theme, and saved cities
- Programmed control algorithms in Python and C++ within the ROS frameworkDesigned multiple custom UI layouts to reflect user settings such as location, theme, and saved cities
- Integrated object detection to recognize signs such as stop signs and trigger stop behavior accordingly
- Developed a vision-based robot using live camera input

Autonomous Robotics Project

ROS, Arduino, OpenCV

- Programmed control algorithms in Python and C++ within the ROS frameworkDesigned multiple custom UI layouts to reflect user settings such as location, theme, and saved cities
- Integrated object detection to recognize signs such as stop signs and trigger stop behavior accordingly
- Developed a vision-based robot using live camera input

AWS Lambda API Project

TypeScript, AWS Lambda, API Gateway

- Created a scalable, serverless backend using AWS Lambda and Express-style middleware
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